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## Fisher Information in Four Models (Problems 1–4)

For each of the following settings, find the following:

- The likelihood
- The log likelihood
- The derivative of the log likelihood
- The expectation of the derivative of the log likelihood
- The variance of the derivative of the log likelihood
- The second derivative of the log likelihood
- The expectation of the second derivative of the log likelihood

Two further items—the variance of the estimator, and its comparison to the variance bound—appear struck through in the posted assignment; since the lecture’s Fisher-information hand-out still asks whether the estimator achieves the bound, each solution closes by noting the comparison where it is immediate.

Throughout,  $Y_1, \dots, Y_n$  are the data,  $y_1, \dots, y_n$  their realizations,  $S = \sum_{i=1}^n Y_i$ , and  $\bar{Y} = S/n$ . The likelihood  $L(\theta)$  is the joint density of the sample evaluated at the realizations and read as a function of the parameter; the derivative  $\ell'(\theta)$  of the log likelihood  $\ell(\theta) = \log L(\theta)$  becomes a random variable when evaluated at the data—the capital/lowercase convention

of Assignment 1 doing real work—and it is that random variable whose moments the assignment requests. Its expectation is zero and its variance equals the negative expected second derivative,

$$\mathbb{E}^\theta\{\ell'(\theta)\} = 0, \quad \text{Var}^\theta\{\ell'(\theta)\} = -\mathbb{E}^\theta\{\ell''(\theta)\} = I_n(\theta),$$

the *Fisher information*, displayed in both forms on the lecture handout (Rabinowitz 2026) and proved as Theorem 8.8.1 of DeGroot and Schervish (pp. 515–516); both identities rest on differentiating  $\int L \, dy_1 \cdots dy_n = 1$  under the integral sign, a regularity step each solution below executes concretely rather than assumes. The course text proves the variance bound  $1/I_n(\theta)$  from the first-derivative form alone—Theorem 1 of HPS (p. 18), by the correlation argument  $\rho^2 \leq 1$  with the quantity  $t = \partial \log L / \partial \theta$  introduced without a name (p. 19)—and never states the second-derivative identity; the two computations the assignment pairs in every setting are exactly the two sides of that missing equation.

## Problem 1

Independent and identically distributed Normal variables with a known variance and unspecified expectation parameter.

SOLUTION:

Let  $Y_1, \dots, Y_n$  be independent and identically distributed  $N(\theta, \sigma^2)$  with  $\sigma^2$  known.

The seven quantities, in the assignment's order:

$$L(\theta) = \prod_{i=1}^n \frac{e^{-(y_i - \theta)^2 / (2\sigma^2)}}{\sqrt{2\pi\sigma^2}} = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_i (y_i - \theta)^2\right] \quad (\text{likelihood})$$

$$\ell(\theta) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (y_i - \theta)^2 \quad (\text{log likelihood})$$

$$\ell'(\theta) = \frac{1}{\sigma^2} \sum_i (Y_i - \theta) = \frac{n}{\sigma^2} (\bar{Y} - \theta) \quad (\text{derivative})$$

$$\mathrm{E}^\theta\{\ell'(\theta)\} = \frac{n}{\sigma^2}(\mathrm{E}^\theta\{\bar{Y}\} - \theta) = \boxed{0} \quad (\text{each } Y_i \text{ has mean } \theta)$$

$$\mathrm{Var}^\theta\{\ell'(\theta)\} = \frac{n^2}{\sigma^4} \mathrm{Var}^\theta(\bar{Y}) = \frac{n^2}{\sigma^4} \cdot \frac{\sigma^2}{n} = \frac{n}{\sigma^2} \quad (\text{independence; scaling})$$

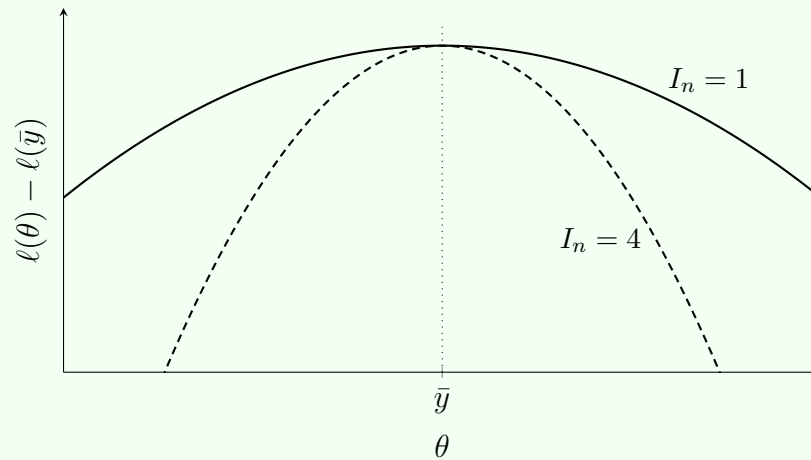
$$\ell''(\theta) = -\frac{n}{\sigma^2} \quad (\text{second derivative})$$

$$\mathrm{E}^\theta\{\ell''(\theta)\} = -\frac{n}{\sigma^2} \quad (\text{expectation of a constant})$$

so that

$$\boxed{I_n(\theta) = \mathrm{Var}^\theta\{\ell'(\theta)\} = -\mathrm{E}^\theta\{\ell''(\theta)\} = \frac{n}{\sigma^2}}$$

Two features of this model are worth naming because the later problems will lose them. The second derivative is not random—the log likelihood is an exact parabola in  $\theta$ , since completing the square gives  $\ell(\theta) = \ell(\bar{y}) - \frac{n}{2\sigma^2}(\theta - \bar{y})^2$ —so the information identity holds without any averaging: observed and expected curvature coincide (Figure 1). And the information does not depend on  $\theta$ : sliding the mean translates the whole family without changing its shape, so every value of the parameter is equally hard to pin down. This is the worked example of both texts (HPS, pp. 17–18, which notes the efficiency of  $\bar{X}$  “is independent of the value of  $\sigma$ ”; DeGroot and Schervish, Example 8.8.3, pp. 516–517). As for the struck comparison:  $\bar{Y}$  is unbiased with  $\mathrm{Var}^\theta(\bar{Y}) = \sigma^2/n = 1/I_n(\theta)$ , so the bound is attained—and the handout’s closing question has the exact answer it hints at, since  $\bar{Y} - \theta = \ell'(\theta)/I_n(\theta)$  identically.



**Figure 1.** INFORMATION IS CURVATURE. The normal log likelihood is an exact parabola,  $\ell(\theta) - \ell(\bar{y}) = -\frac{1}{2}I_n(\theta - \bar{y})^2$ , drawn for  $I_n = n/\sigma^2$  equal to 1 (solid) and 4 (dashed). More information means a sharper peak: the data argue harder against parameter values away from  $\bar{y}$ . In the models of Problems 2–4 the curvature varies from sample to sample and only its expectation equals the information.



## Problem 2

Independent and identically distributed Poisson variables with an unknown expectation parameter.

SOLUTION:

Let  $Y_1, \dots, Y_n$  be independent and identically distributed  $\text{Poisson}(\theta)$ ,  $\theta > 0$ , so that each has pmf  $e^{-\theta}\theta^y/y!$  and mean and variance both  $\theta$ . With  $s = \sum_i y_i$ :

$$L(\theta) = \prod_{i=1}^n \frac{e^{-\theta}\theta^{y_i}}{y_i!} = \frac{e^{-n\theta}\theta^s}{\prod_i y_i!} \quad (\text{likelihood})$$

$$\ell(\theta) = -n\theta + s \log \theta - \sum_i \log(y_i!) \quad (\text{log likelihood})$$

$$\ell'(\theta) = \frac{S}{\theta} - n \quad (\text{derivative})$$

$$\mathbb{E}^\theta\{\ell'(\theta)\} = \frac{n\theta}{\theta} - n = \boxed{0} \quad (\text{Poisson mean; linearity})$$

$$\text{Var}^\theta\{\ell'(\theta)\} = \frac{\text{Var}^\theta(S)}{\theta^2} = \frac{n\theta}{\theta^2} = \frac{n}{\theta} \quad (\text{independence; Poisson variance = mean})$$

$$\ell''(\theta) = -\frac{S}{\theta^2} \quad (\text{second derivative})$$

$$\mathbb{E}^\theta\{\ell''(\theta)\} = -\frac{n\theta}{\theta^2} = -\frac{n}{\theta} \quad (\text{Poisson mean again})$$

so that

$$\boxed{I_n(\theta) = \text{Var}^\theta\{\ell'(\theta)\} = -\mathbb{E}^\theta\{\ell''(\theta)\} = \frac{n}{\theta}}$$

Unlike Problem 1, the second derivative  $-S/\theta^2$  is now genuinely random—each sample carries its own curvature—and the two routes to the information no longer coincide termwise but only in expectation, which is what the identity is for. The information  $n/\theta$  falls as  $\theta$  grows: a Poisson distribution with a large mean is also a spread-out one (variance  $\theta$ ), so each observation pins the parameter down less. The comparison the

assignment struck is again immediate:  $\bar{Y}$  is unbiased with  $\text{Var}^\theta(\bar{Y}) = \theta/n = 1/I_n(\theta)$ , attaining the bound, and indeed  $\bar{Y} - \theta = (\theta/n) \ell'(\theta)$  is linear in the derivative of the log likelihood—the equality condition in DeGroot and Schervish’s form of the inequality, and precisely how their Example 8.8.9 (pp. 521–522) certifies this estimator. The model is absent from HPS’s own examples in §2.2, which run the normal and the Bernoulli only (pp. 17–21). ■



### Problem 3

Independent and identically distributed Exponential random variables with unknown expectation parameter.

SOLUTION:

The expectation is the parameter, so write the density as  $f(y \mid \theta) = \theta^{-1}e^{-y/\theta}$  for  $y > 0$ , under which  $E^\theta\{Y_i\} = \theta$  and  $\text{Var}^\theta(Y_i) = \theta^2$ . Then:

$$L(\theta) = \prod_{i=1}^n \theta^{-1} e^{-y_i/\theta} = \theta^{-n} e^{-s/\theta} \quad (\text{likelihood})$$

$$\ell(\theta) = -n \log \theta - \frac{s}{\theta} \quad (\text{log likelihood})$$

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{S}{\theta^2} \quad (\text{derivative})$$

$$E^\theta\{\ell'(\theta)\} = -\frac{n}{\theta} + \frac{n\theta}{\theta^2} = \boxed{0} \quad (\text{exponential mean; linearity})$$

$$\text{Var}^\theta\{\ell'(\theta)\} = \frac{\text{Var}^\theta(S)}{\theta^4} = \frac{n\theta^2}{\theta^4} = \frac{n}{\theta^2} \quad (\text{independence; exponential variance } \theta^2)$$

$$\ell''(\theta) = \frac{n}{\theta^2} - \frac{2S}{\theta^3} \quad (\text{second derivative})$$

$$E^\theta\{\ell''(\theta)\} = \frac{n}{\theta^2} - \frac{2n\theta}{\theta^3} = -\frac{n}{\theta^2} \quad (\text{exponential mean again})$$

so that

$$\boxed{I_n(\theta) = \text{Var}^\theta\{\ell'(\theta)\} = -E^\theta\{\ell''(\theta)\} = \frac{n}{\theta^2}}$$

The second derivative is again random, and this time *both* of its terms matter to the expectation: the positive  $n/\theta^2$  alone would say the log likelihood curves upward, and only after averaging the random  $-2S/\theta^3$  does the concavity  $-n/\theta^2$  emerge. The struck comparison:  $\bar{Y}$  is unbiased for  $\theta$  with  $\text{Var}^\theta(\bar{Y}) = \theta^2/n = 1/I_n(\theta)$ —attained, via  $\bar{Y} - \theta = (\theta^2/n)\ell'(\theta)$  once more. The parameterization deserves the emphasis the

assignment gives it (“unknown *expectation* parameter”), because the same model told in the rate parameterization  $\beta = 1/\theta$  behaves differently: DeGroot and Schervish’s Example 8.8.6 (p. 520) works that version, finds  $I_n(\beta) = n/\beta^2$ , and shows the natural unbiased estimator of  $\beta$ , namely  $(n - 1)/S$ , has variance  $\beta^2/(n - 2)$ —*strictly above* the bound  $\beta^2/n$ , no unbiased estimator of the rate attaining it. Estimating the mean of an exponential is a bound-attaining problem; estimating its rate is not; and the difference is invisible until the two computations are set side by side. ■

## Problem 4

Independent and identically distributed random variables with a density of the form

$$f(y) \exp[\theta y - \psi(\theta)],$$

where  $f$  is a specified density,  $\theta$  is the unknown parameter, and  $\psi$  is the function of  $\theta$  that makes the expression integrate to 1.

SOLUTION:

The normalization that defines  $\psi$  is the engine of everything below, so state it first:

$$\int f(y) e^{\theta y - \psi(\theta)} dy = 1 \quad \text{for every } \theta, \quad (*)$$

with sums replacing integrals when  $f$  is a pmf. The seven quantities:

$$L(\theta) = \prod_{i=1}^n f(y_i) e^{\theta y_i - \psi(\theta)} = \left[ \prod_i f(y_i) \right] \exp [\theta s - n\psi(\theta)] \quad (\text{likelihood})$$

$$\ell(\theta) = \sum_i \log f(y_i) + \theta s - n\psi(\theta) \quad (\text{log likelihood})$$

$$\ell'(\theta) = S - n\psi'(\theta) \quad (\text{derivative})$$

The expectation of the derivative requires the mean of  $Y$ , and  $(*)$  supplies it: differentiating both sides in  $\theta$ —the regularity step, legitimate because the integrand is smooth and exponentially dominated in  $\theta$ —gives

$$\int f(y) e^{\theta y - \psi(\theta)} (y - \psi'(\theta)) dy = 0, \quad \text{i.e.} \quad E^\theta\{Y\} = \psi'(\theta),$$

and differentiating once more gives

$$\int f(y) e^{\theta y - \psi(\theta)} (y - \psi'(\theta))^2 dy - \psi''(\theta) = 0, \quad \text{i.e.} \quad \text{Var}^\theta(Y) = \psi''(\theta) :$$

the normalizer generates the moments, mean by its slope and variance by its curvature.

With these,

$$\mathbb{E}^\theta \{\ell'(\theta)\} = n\psi'(\theta) - n\psi'(\theta) = \boxed{0} \quad (\text{mean } \psi'; \text{ linearity})$$

$$\text{Var}^\theta \{\ell'(\theta)\} = \text{Var}^\theta(S) = n\psi''(\theta) \quad (\text{independence; variance } \psi'')$$

$$\ell''(\theta) = -n\psi''(\theta) \quad (\text{second derivative})$$

$$\mathbb{E}^\theta \{\ell''(\theta)\} = -n\psi''(\theta) \quad (\text{expectation of a constant})$$

so that

$$\boxed{I_n(\theta) = \text{Var}^\theta \{\ell'(\theta)\} = -\mathbb{E}^\theta \{\ell''(\theta)\} = n\psi''(\theta)}$$

Like the normal—and for the same reason, a log likelihood whose  $\theta$ -dependence is built from  $\theta s$  and  $-n\psi(\theta)$  alone—the second derivative is nonrandom, so the information identity holds without averaging. And the two differentiations of (\*) are not a trick special to this family: they are the general proofs that the score has mean zero and that its variance equals the negative expected curvature, run in a family clean enough to show every step. As for attainment:  $\bar{Y}$  is unbiased for the *mean*  $m(\theta) = \psi'(\theta)$  rather than for  $\theta$  itself, so the relevant bound is DeGroot and Schervish's general form  $[m'(\theta)]^2 / I_n(\theta)$  (Thm. 8.8.3, pp. 518–519), which here equals  $[\psi''(\theta)]^2 / (n\psi''(\theta)) = \psi''(\theta)/n = \text{Var}^\theta(\bar{Y})$ : attained, because  $\bar{Y} - \psi'(\theta) = \ell'(\theta)/n$  is linear in the derivative of the log likelihood (their Exercise 11 of §8.8, p. 527, asks for exactly this).

The family is not a fourth example so much as the first three seen whole. Each earlier model is the display above after a change of variable: the normal with known  $\sigma^2$  has natural parameter  $\theta = \mu/\sigma^2$  with  $\psi(\theta) = \sigma^2\theta^2/2$ ; the Poisson has  $\theta = \log \lambda$  with  $\psi(\theta) = e^\theta$  (its base  $f$  a pmf—the density-for-both convention of HPS, p. 2, earning its keep); the exponential has  $\theta = -1/\mu$  with  $\psi(\theta) = -\log(-\theta)$ . In the natural parameter the informations are  $n\sigma^2$ ,  $ne^\theta = n\lambda$ , and  $n/\theta^2 = n\mu^2$ —not the  $n/\sigma^2$ ,  $n/\lambda$ ,  $n/\mu^2$  of Problems 1–3, and the discrepancy is the lesson: information is attached to a parameterization, not to a model, transforming by the square of the reparameterization derivative,  $I(\eta) = I(\theta) (d\theta/d\eta)^2$ . Applying that rule to each pair above reproduces the earlier answers exactly, which is as good a check on all four problems as any. ■

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